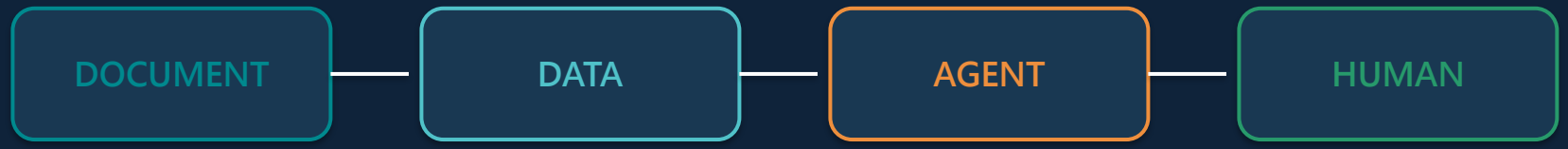
# Smart Warehouse AI Agent

## 智慧倉儲單據辨識與庫存異常決策代理人

### Final Project Proposal

AI Agent × Domain Knowledge × Data Analytics × Real-World Application




組員 / 課程 / 指導教師：請於正式繳交前填寫 | 2026-10-31

# Research Motivation | 研究動機



INTRODUCTION

**87%**
使用 WMS

**51%**
仍人工蒐集生產力資料

**76%**
使用 inventory accuracy 指標

## 數位化 ≠ 資料可靠 ≠ 異常可解釋

- 前端單據仍可能含 OCR、缺欄、單位與版型錯誤。
- 決策需要文件、庫存、規則、證據與人工責任共同成立。

Source: Michel (2025), Warehouse/DC Operations Survey, n=101；產業調查，非全球母體推論。

# Problem Definition｜從文件到安全決策



INTRODUCTION

**INPUT**
圖片 / PDF
庫存 snapshot
主檔 / 歷史

**UNCERTAINTY**
版型 / OCR error
缺欄 / 資料衝突
過期 snapshot

**DOMAIN CHECK**
duplicate
數量 / 單位
倉別 / SKU

**OUTPUT**
欄位 + 異常
建議 + evidence
human gate

**研究單位不是「辨識多少字」，而是「證據是否足以支持可追溯的倉儲建議」。**

Scope: recommendation-only；不直接寫正式 WMS/ERP。

# Research Questions & Measurable Objectives



INTRODUCTION

| | |
|---|---|
| RQ1 文件理解 | CER / Field EM / Document EM<br>可重現、可稽核、學生電腦可執行 |
| RQ2 異常決策 | Precision / Recall / macro-F1<br>可重現、可稽核、學生電腦可執行 |
| RQ3 Agent 價值 | 工具選擇 / 完成率 / 人工介入<br>可重現、可稽核、學生電腦可執行 |
| RQ4 Trustworthiness | evidence coverage / false auto-accept<br>可重現、可稽核、學生電腦可執行 |

RQ 與指標需在 test 前凍結；正式實驗目前 Not Run。

# Related Work | 10 篇核心研究與 Research Gap



RELATED WORK

| Ref | Year | Problem / Method | Domain / Dataset | Finding / Limitation |
|---|---|---|---|---|
| [1] DocAgent | 2025 | multi-agent retrieval + reviewer | 2 long-doc benchmarks | 支援交叉核對；非倉儲 |
| [2] AID-Agent | 2025 | OCR + LLM agent + tools | 44 industrial PDFs | 工具化抽取；小型案例 |
| [3] Domain IE | 2025 | human/rules/LLM comparison | real-estate case | 存在成本與維護權衡 |
| [4][5] SROIE/DocVQA | 2019/21 | OCR/KIE/VQA benchmarks | receipts/documents | 無庫存與決策標註 |
| [6] IoT review | 2025 | systematic review | warehouse/inventory IoT | 實務導入仍有限 |
| [7] Inventory anomaly | 2025 | ensemble（摘要） | 12,358 SKUs（摘要） | F1 87.3%（摘要）；全文待查 |
| [8] AI picking | 2025 | case + simulation + interviews | warehouse picking | 效益與導入挑戰並存 |
| [9][13] Smart warehouse/DT | 2026 | SLR / review framework | warehouse ecosystems | 整合視角；非 document agent |

**GAP 文件 AI 多停在抽取 / 問答；倉儲 AI 多假設資料已結構化。**
**OUR PROJECT document → inventory context → domain rules → evidence-backed decision → human gate**

AI: [1]–[5]；Warehouse: [6]–[9], [13]。詳細八欄表見 README/proposal.md。

# Five Expert Mindsets｜領域知識如何進入設計



DEEP RESEARCH

**1**

流程與狀態守恆

before → event → after

**2**

唯一識別與追溯

ID + provenance

**3**

Fitness for Use

品質依任務與風險

**4**

例外與風險優先

把人工留給高風險

**5**

閉環校正與根因

異常 ≠ 根因

**Design translation：snapshot freshness | evidence IDs | quality gates | human escalation | append-only audit**

Evidence: GS1 [10], NIST [12], DeHoratius & Raman [14], Destro [DR16], Chuang & Oliva [DR20].

# Three Professional Controversies



DEEP RESEARCH

| | | |
|---|---|---|
| 高度自動化 | Human-in-the-loop | CURRENT CONSENSUS<br>Risk-based automation |
| Deterministic Rules | Learning Agent | CURRENT CONSENSUS<br>Hybrid responsibility boundary |
| Real operational data | Synthetic / privacy data | CURRENT CONSENSUS<br>Mixed evidence + honest validity |

**Unresolved：threshold、人工負荷、domain shift、distribution gap、成本效益。**

Consensus is evidence synthesis, not a claim that every study agrees. Thresholds remain local empirical questions.

# Ten Deep Questions｜以失敗情境檢查理解



DEEP RESEARCH

**Q01**
高 OCR accuracy 為何仍會錯決策？

**Q02**
文件與 snapshot 時間不同怎麼辦？

**Q03**
自動通過與危險錯放如何權衡？

**Q04**
OCR、barcode、主檔衝突時信誰？

**Q05**
什麼是充分 evidence？

**Q06**
為何保留 OCR + Rules baseline？

**Q07**
為何 planner 必須 bounded？

**Q08**
人工核准後發現錯誤怎麼追？

**Q09**
新版型低信心時能降 threshold 嗎？

**Q10**
盤點差異為何不能直接定因？

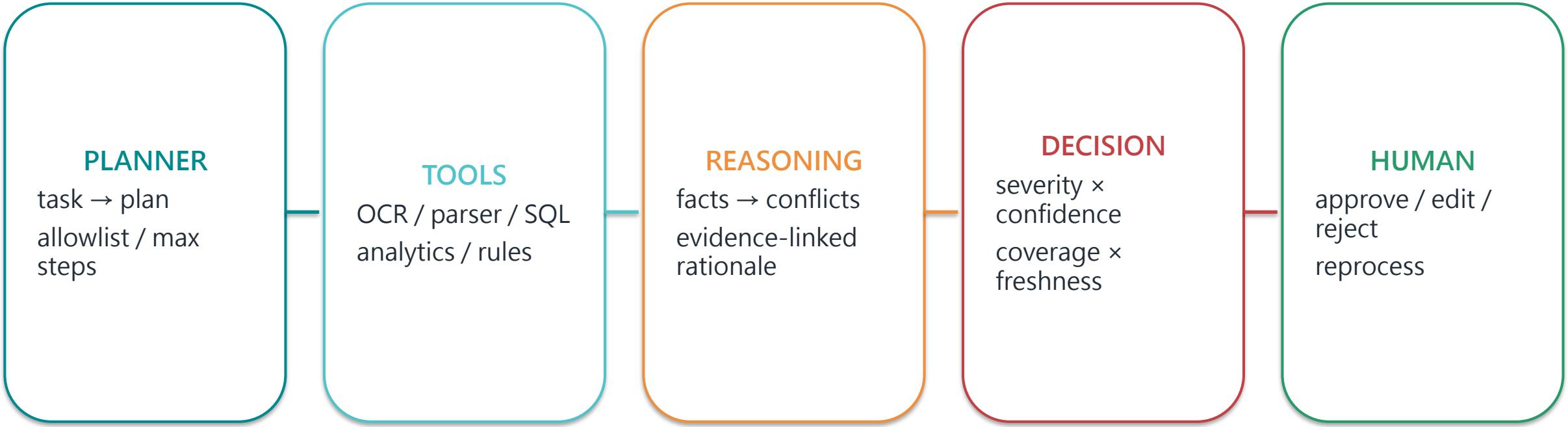
**核心原則：觀測、推論、未知與人工處置必須分開保存。**

完整分析、證據與設計啟示見 deep_research.md Q3。

# Proposed Domain AI Agent｜能力與責任邊界



PROPOSED SCHEME



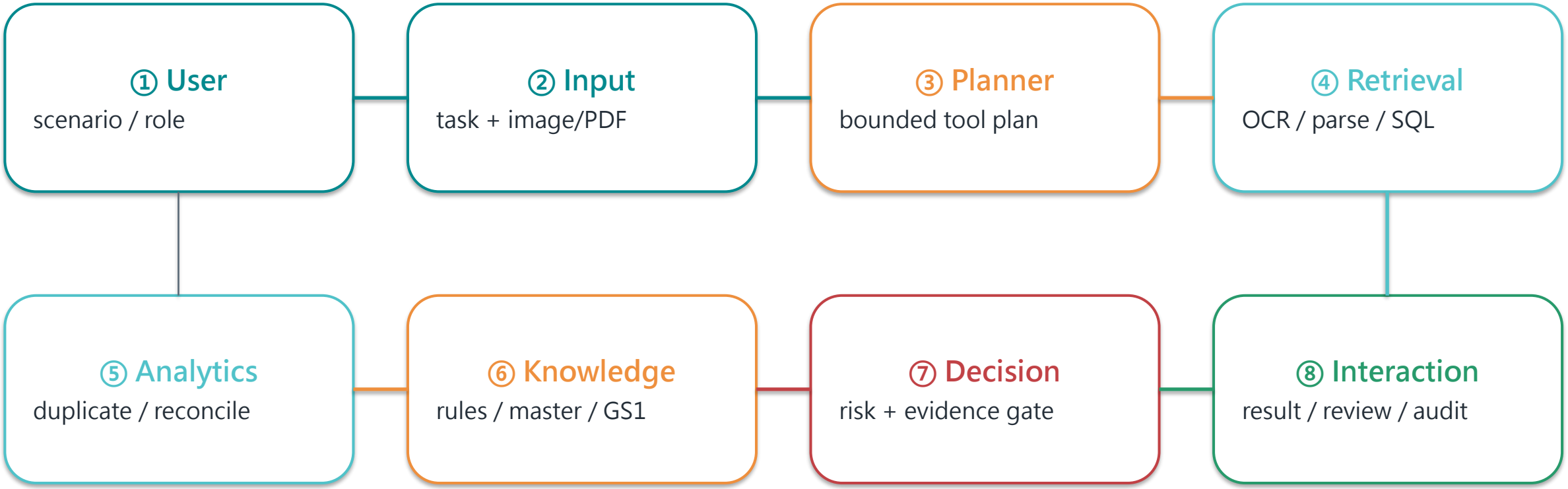

- LLM：任務映射與受控工具選擇；不得創造數量、SKU 或政策。
- Rules / SQL / statistics：可重現計算；最終安全 gate 與正式寫入權限不交給 LLM。

Agent ≠ fixed pipeline；只有能依 state 選工具、補查、停止或升級者才稱 Agent。

# System Architecture | ①–⑧ 可測試模組



PROPOSED SCHEME



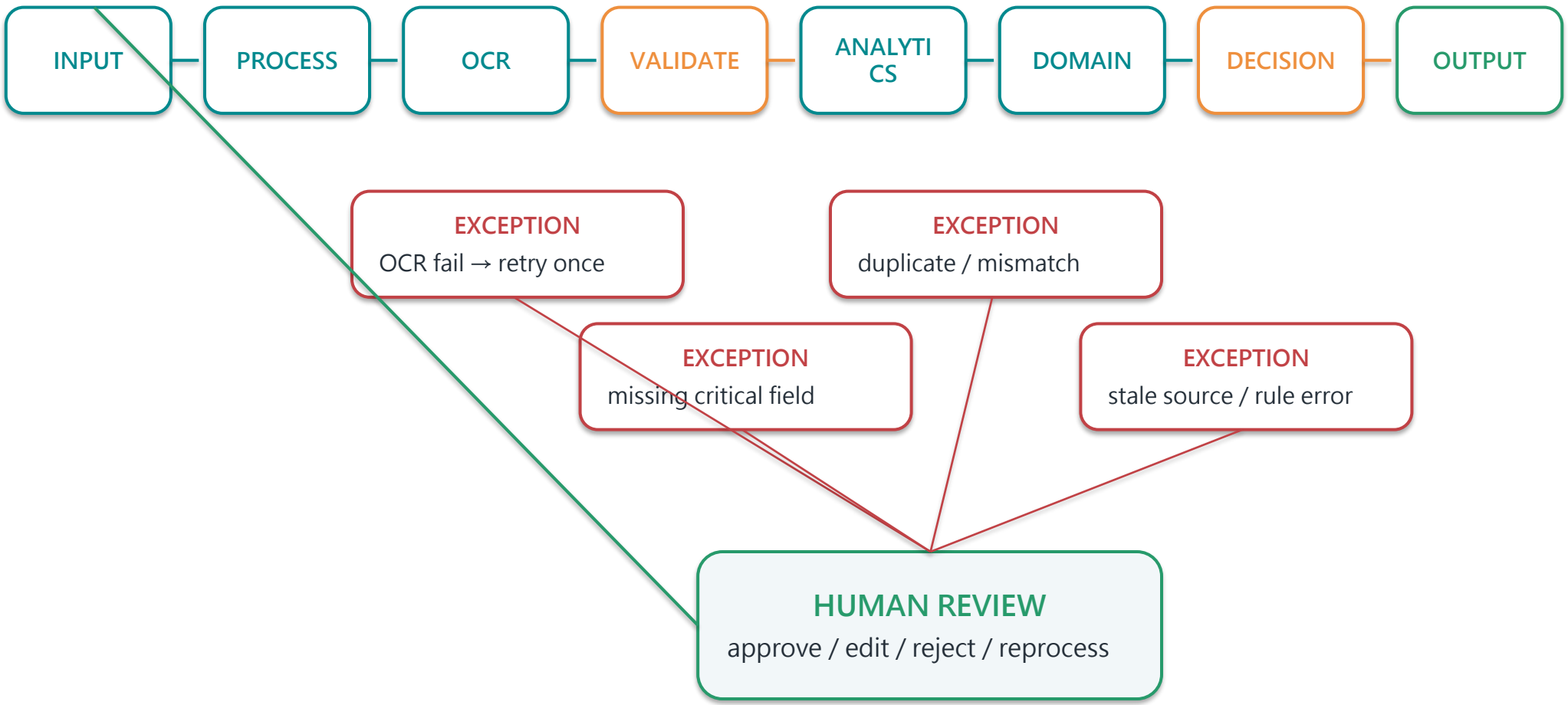

Human reprocess loop returns to Planner with the same root trace; no formal WMS write in MVP.

Target design: local-first, read-only inventory, evidence-first, human-gated, append-only audit.

# Flowchart | Normal Path, Exception, Human Loop



PROPOSED SCHEME




**Fail-safe：來源不可用 ≠ 查無資料；模糊候選 ≠ confirmed duplicate；reprocess 不覆寫舊 trace。**

Normal=teal；Decision=orange；Exception=red；Human interaction=green.

# Algorithm 1 | Bounded Planning + Evidence Gate



PROPOSED SCHEME

```
01  validate input; start append-only trace
04  task ← UNDERSTAND_TASK(q, allowed scenarios)
09  plan ← BUILD_BOUNDED_PLAN(allowed tools, max steps)
15  OCR → retry once → ESCALATE if unusable
24  parse typed schema; link fields to OCR spans
33  read-only inventory / history / master queries
43  exact duplicate ≠ fuzzy candidates
50  evaluate versioned warehouse rules
57  reason from evidence only
62  confidence + coverage + risk
67  duplicate / mismatch / uncertainty → HUMAN REVIEW
79  ASSERT_NO_WMS_OR_INVENTORY_WRITE_OCCURRED
91  append trace and return
```

**INVARIANT 1**

All tools are allowlisted and schema-valid.

**INVARIANT 2**

Every material claim has evidence ID.

**INVARIANT 3**

High risk / uncertainty cannot auto-write.

Complete 92-line pseudocode: algorithm.md ; slide shows critical control points.

# MVP Implementation | What Exists vs. What Is Planned



TECHNICAL FEASIBILITY

## IMPLEMENTED

- Streamlit upload
- mock/sidecar OCR
- typed parser
- SQLite seed/query
- exact duplicate + rules
- human confirm + audit

## PARTIAL

- bounded planner
- embedded-text PDF
- confidence gate
- target architecture mapping

## PLANNED

- real/scanned OCR benchmark
- LLM adaptive planner
- fuzzy duplicate
- Google Sheets adapter
- formal WMS integration

**Python 3.11 + Streamlit + SQLite + Pandas | 一般學生電腦，不訓練大型模型。**

Automated tests: 6/6 synthetic fixtures passed; this is not formal experiment evidence.

# Dataset Design | No Leakage, No Sensitive Documents



EXPERIMENTS

## PUBLIC PRETEST

SROIE / DocVQA
pipeline sanity only

## MAIN DATASET

anonymous / synthetic
Chinese + English warehouse docs

## PAIRED CONTEXT

immutable snapshots
master + history + gold decision

## GROUP SPLIT

vendor_id + template_id

60 / 20 / 20
seed = 20261024

Same original + augmentation
MUST stay in one split

Actual sample size / license / class distribution：NOT RUN / 待查證

Main dataset target is planned, not collected: 300–500 documents, ≥3 types, ≥5 templates.

# Three Baselines | Fair Paired Comparison



EXPERIMENTS

## B0 MANUAL

人工輸入 + 人工核對

Operational reference

## B1 FIXED

OCR + Fixed Rules

Non-agent control

## B2 AGENT

OCR + Domain AI Agent

Proposed method

**COMMON：OCR | parser | rules | thresholds | hardware | test manifest**

- B1 固定順序、不 replan；B2 才能依中間 state 選工具或升級。
- 若 B1/B2 使用不同 OCR 或規則，就無法把差異歸因於 Agent planning。

All baselines use the same test documents and immutable database snapshots.

# Evaluation Metrics & Procedure



EXPERIMENTS

**DOCUMENT**
CER / Character Accuracy
Field EM / Document EM

**ANOMALY**
Precision / Recall / macro-F1
Duplicate Accuracy + F1

**OPERATIONS**
Mean / median / P95 time
Human intervention

**AGENT**
Tool precision / recall
Safe tool / task completion

1 Annotate + adjudicate → 2 Freeze split/config → 3 Run paired B0/B1/B2 → 4 Bootstrap 95% CI → 5 Error analysis

| Metric | B0 | B1 | B2 | Status |
|---|---|---|---|---|
| macro-F1 / time / intervention / completion | — | — | — | NOT RUN |

Primary endpoint proposed: anomaly macro-F1（teacher confirmation required）; all results Not Run.

# Expected Contributions & Feasibility



CONCLUSION

## C1
Research framing

OCR → domain decision agent

## C2
System design

evidence-first + human-gated

## C3
Evaluation

fair B0/B1/B2 + safety metrics

## C4
Artifact

student-computer open-source MVP

**EXPECTED：B2 可能改善跨來源異常處理與 auditability；代價可能是 latency、complexity 與人工 threshold 調整。**

**NOT CLAIMED：尚無正式 accuracy、F1、time saving、ROI 或「優於所有方法」的結果。**

Expected outcomes are hypotheses to test, not observed findings.

# References, Submission Status & Next Steps



CONCLUSION

[1] Sun et al. (2025), DocAgent, EMNLP. doi:10.18653/v1/2025.emnlp-main.893
[2] Li, Conen & Aller (2025), AID-Agent, REALM. doi:10.18653/v1/2025.realm-1.6
[3] Principe et al. (2025), LDK, ACL Anthology 2025.ldk-1.25
[6] Puga Mendoza et al. (2025), Procedia CS. doi:10.1016/j.procs.2025.01.185
[8] Fakhrai Rad et al. (2025), Review of Managerial Science. doi:10.1007/s11846-025-00858-1
[9] El Kadmiri et al. (2026), Cleaner Logistics and Supply Chain. doi:10.1016/j.clscn.2026.100340
[10] GS1 Logistic Label Guideline 1.3 (2019) | [12] NIST AI RMF 1.0 (2023)
[13] Alam et al. (2026), ICT Express. doi:10.1016/j.icte.2026.01.009
Full verified list: README.md / proposal.md / deep_research.md

**NOW**
Proposal + Design + MVP

**NEXT**
teacher confirms protocol

**PILOT**
30–50 compliant cases

**FREEZE**
manifest + versions

**RUN**
B0 / B1 / B2

Repository: https://github.com/john19991225-commits/-